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REGULAR PAPER

CDGRS-LOD model: leveraging linked open data in collaborative cross-domain group recommender system

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Abstract
This study introduces a novel Cross-Domain Group Recommender System leveraging Linked Open Data (LOD) (CDGRS-LOD), designed to improve the accuracy and relevance of group recommendations. Group recommender systems (GRS) are increasingly used to deliver collective recommendations, but their effectiveness is often constrained by sparse user data and limited semantic context during group formation. The proposed CDGRS-LOD model enriches movie metadata with structured external knowledge, incorporating cross-domain attributes such as the Music Composer feature extracted from DBpedia via SPARQL queries. This enhancement improves the model's understanding of relationships between users and items, thereby enabling more accurate and contextually relevant group recommendations. Singular Value Decomposition (SVD) predicts missing ratings from the enriched metadata, effectively mitigating sparsity. Two sparsity reduction levels, 1% and 5%, were evaluated, with the 5% reduction yielding the best performance and subsequently used for group formation with a k-Nearest Neighbours (kNN) model. The model is evaluated against existing methods, including GRS-LOD, Archi2_bkm, Archi2_km, and Predict & Cluster. Results show that CDGRS-LOD achieves a Root Mean Square Error (RMSE) of 0.5917, outperforming GRS-LOD (0.7816) and traditional approaches (0.9435–0.9781), representing up to 24% improvement over the best baseline. These findings confirm that integrating cross-domain knowledge with LOD enhances metadata richness, reduces sparsity, lowers prediction errors, and improves group recommendation quality.

Keywords Cross-domain group recommender system · Data sparsity · Linked open data · Semantic

1 Introduction

Recommender systems (RS) have become a prominent solution for delivering relevant information and have been successfully applied across various domains. It is integral in numerous online platforms presently. With the exponential growth of digital content, users often face information overload, making it challenging to identify items that match their interests. RS address this challenge by filtering and personalizing information to enhance user satisfaction and decision-making. The systems enable users to navigate vast information and discover relevant items according to their preferences. Consequently, RS are widely implemented across domains, such as e-commerce [1–3], health [4, 5], music [6, 7] and education [8]. Commonly, RS rely on several techniques to suggest items to users based on historical data, behavioural patterns, or item characteristics, including collaborative and content-based filtering. Collaborative filtering (CF) employs user-item interactions to identify preference patterns among similar users [9]. Meanwhile, content-based filtering applies item attributes to recommend similar items to those the user has previously liked [10].

Although popular, traditional RS face significant challenges, particularly when presented with notable user-item interaction data sparsity, which can lead to inaccurate or irrelevant recommendations. Accordingly, most studies utilise cross-domain [2, 11, 12] and linked open data (LOD) technology [13–15] to overcome the matter in individual-based

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RS. Meanwhile, numerous RS either focus on individual recommendations in various study contexts, including session context in e-commerce domain [3], knowledge graph-based in human-centred RS [16], conditional diffusion in generative recommendation [17], and hybrid RS with fuzzy [18]. Although the models demonstrated potential in addressing sparsity at the individual level, they frequently fail to adequately overcome the complexities involved in group settings in group recommender systems (GRS) [19–21], primarily due to incomplete data. Beyond individual personalisation, RS have increasingly evolved to support group-based recommendations, requiring more sophisticated algorithms capable of balancing diverse preferences within a collective decision-making context. Consequently, more sophisticated approaches are necessary to effectively balance diverse preferences and deliver accurate recommendations that resonate with group decision-making contexts.

Current GRS reports addressed data sparsity issues following group formation [22], denoting that the matter is encountered between groups. Nevertheless, resolving data sparsity before forming groups would be more efficient, such as at the individual user level. Several approaches have been proposed for solving data sparsity, including recursive filtering [23], artificial neural network [24, 25], and item metadata embedding [26] Nevertheless, the issue of data sparsity remains a challenge in GRS.

This study developed a novel model, the cross-domain GRS leveraging LOD (CDGRS-LOD) to enhance group recommendation capabilities, which integrates the LOD technology with cross-domain recommendation techniques. The proposed model employs the rich, structured data provided by LOD in cross-domain RS to improve the accuracy and relevance of recommendations in group settings. Consequently, the model can access diverse data sources, enriching user profiles and item representations, which are crucial for making informed recommendations.

Our study postulated that reducing data sparsity and enhancing the quality of data input can lead to considerably improved recommendation performance. The proposed CDGRS-LOD model also may bridge the gap between traditional recommendation techniques and the complexities of group recommendations by effectively applying external knowledge and cross-domain insights. Furthermore, the model was developed to provide tailored recommendations reflecting the collective interests of all group members by focusing on the interplay between individual user preferences and group dynamics.

This study aimed to address challenges, such as data sparsity and diverse user preferences in group settings. Cross-domain feature from movie and music domains were integrated in the model, which were extracted from DBpedia with SPARQL queries. Parallel processing was also employed to efficiently query large-scale LOD repositories with multi-core processors, significantly elevating the performance of the system.

The model proposed in this study enriches movie metadata with structured external knowledge using LOD-based integration. This allows a more nuanced understanding of relationships between users and items. The contributions of this study are summarised as follows:

- A CDGRS-LOD model applying the cross-domain approach with LOD technology to enrich item information for use in a collaborative GRS was proposed.
- An approach utilising similarity-based DBpedia attributes ('dbo:director' and 'dbo:starring') for movie domain and 'dbo:musicComposer' for music domain was suggested.
- The employment of cross-domain approach with LOD technology in group formation enhanced the quality of recommendations within groups by alleviating sparsity.
- The impact of sparsity reduction was examined by comparing the model's performance at 1% and 5% sparsity levels.

Section 2 of this article outlines related studies, while the CDGRS-LOD model is detailed in Section 3. In Section 4, the experimental setup utilised is discussed. The outcomes are documented in Section 5, and Section 6 concludes this article.

2 Related works

Although Khan et al. [27] reviewed studies on cross-domain recommender systems (CDRS), the discussion of linked open data (LOD) methods and their integration within cross-domain frameworks was limited, with a primary focus on individual recommender systems (RS). The present study aims to extend this discussion by examining CDRS in greater depth, centring on the role of LOD technology and its utilisation within cross-domain approaches.

A notable finding of this study is the potential gap between existing studies and the application of cross-domain and LOD methods in group recommender systems (GRS). Accordingly, this section discusses the essential concepts underlying individual RS, GRS, CDRS, and LOD technologies, providing a foundation for the subsequent analysis and identification of research gaps.

2.1 Cross-domain recommender system

The CDRS were developed to overcome data sparsity issues and have been gaining prominence. The systems transfer knowledge from a source domain consisting of considerable data to a destination domain with limited information [28].

Consequently, the CDRS enhance the performance of recommendations for the target domain. Generally, the CDRS enhance the performance of the target domain recommendations.

The CDRS can be built through numerous approaches. According to Li and Tuzhilin [29], extensions of single-domain recommendation models are the most common CDRS. Zhu et al. [30] indicated that CDRS have different scenarios for target domain recommendations, as illustrated in Fig. 1, including single-, dual-, and multi-targets, and multi-domain. Conventional single-target focuses on alleviating a sparse target domain, aiming to provide recommendations in a specific domain in two domain scenarios. Meanwhile, dual-target improves both domains and offers superior recommendations in each domain.

The accuracy of each CDRS in each domain is increased simultaneously utilising the auxiliary knowledge across domains in multi-target scenarios, which involve over three domains. Users and items also overlap across the multiple domains. A collection of items across various domains is recommended to a set of users in different domains by applying auxiliary knowledge across domains. Multi-domain and dual- and multi-target CDRS are generally more sophisticated in model designs and processes.

Based on Fig. 1, cross-domain data can be categorised from three perspectives: user-, item-, and content-level relevance [30]. Data relevance identification between domains contributes to knowledge transfer to the target domain. The information can be based on data overlapping, resulting in more straightforward mapping between domains with common features, users, or items.

Li and Tuzhilin [29] argued that certain CDRS models fail to adequately address the relationship between different domains. The models do not simultaneously improve the effectiveness of recommendations in both domains. Furthermore, the models might restrict the applicability of cross-domain user interaction data. Existing studies on CDRS with diverse methods and algorithms to overcome individual RS issues applied various evaluation metrics on real-world domains and datasets [2, 11, 31–33]. Xuan et al. [34] also performed comparison analysis regarding CDRS, offering an extensive review of the impact of LOD and CDRS in enhancing recommendation efficacy for both individual and group users. They additionally emphasised ongoing challenges, including data sparsity and cold-start issues within GRS.

2.2 Linked open data

The LOD technology is the foundation for creating a knowledge graph applicable in CDRS with a semantic web core. Consequently, LOD technology is a component of the semantic web, and its datasets create new difficulties and issues for complicated web applications and next-generation RS [35]. Meanwhile, linked resource description framework (RDF) statements are contained in a massive, decentralised knowledge repository [36]. Metadata is described and exchanged through RDF statements, allowing a standardised data interchange based on relationships.

Multiple data sources are combined utilising RDF. The LOD cloud includes structured data on various subjects, including biological sciences, media (such as books and movies), and political and geographical information. Typically, the data are accessed through DBpedia, which is the Wikipedia of RDF mapping. The instrument is widely acknowledged as the development epicentre of Web of Data [36]. Data mapping is crucial in CDRS that employs LOD technology.

Pereira [37] applied LOD in DBpedia to diminish prediction errors in RS collaborative filtering. The empirical study by Natarajan et al. [38] integrated LOD technology in individual RS. Other studies incorporated LOD or semantic and cross-domain approaches in individual RS [39]. Nonetheless, only one study applied LOD technology to solve issues in introducing LOD in GRS [40]. The study did not utilise a cross-domain approach.

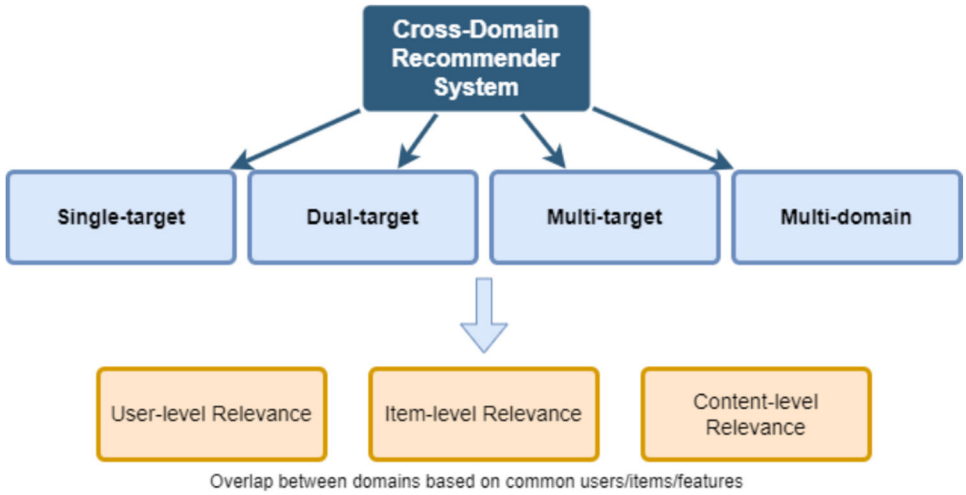
2.3 Group recommender system

Learning algorithms are applicable to RS in at least three primary ways: collaborative, content-based, and hybrid filtering. The approaches can also be implemented in GRS. Each strategy is employed across various platforms, offering benefits and drawbacks. The recommendation process also alters when targeting a group of users instead of an individual. In such cases, a group profile representing the combination of the preferences of all group members is formed. The profile is influenced by the individual preferences of each user and the overall composition of the group [41].

The GRS is operated in two stages; the creation of a group profile and the recommendation of products tailored to the group. Based on the perspective of group profile construction, GRS can be classified into stable groups, where group composition remains consistent, and random groups, where group members may vary, regardless of the application domain [42]. Wang et al. [43] stated that stable group members can add their preferences to the group profile, and over time, the preferences of the group may become centralized. Conversely, members in random groups have no opportunity to express their opinions and have notably contradictory group preferences [43].

The GRS and individual RS face several issues. For instance, few and insufficient user-item ratings lead to GRS and individual RS matters, resulting in data sparsity. Liang et al. [41] concluded that data sparsity is another problematic

Fig. 1 Type of CDRS



issue in GRS. The issue is worse in GRS as the historical records of groups are considerably smaller than those of individuals. Consequently, the quality of recommendations offered to a group suffers from a lack of data. Insufficient information arises from ineffective group creation, which frequently includes individuals with sparse user profiles.

[19, 44–46] presented study on GRS utilising other than LOD technology and cross-domain approach. While Sun et al. [47] performed studies on CDRS for groups with diverse algorithms to address GRS-related issues.

2.3.1 Evaluating group recommender system

According to Trattner et al. [48], the two evaluation types in GRS are "offline" and "online". "Offline" assessments employ information from train and test sets. From the machine learning viewpoint, a dataset is commonly divided into 20% and 80% test and training sets, respectively. A test set is created to ensure the reliable evaluations of algorithms with untrained data. Meanwhile, "online" assessments are frequently performed on a whole system to determine user feedback.

Evaluation metrics are determined according to the nature of the task and the system objectives. Classification, error, and ranking are the three common evaluation metrics in "offline" evaluations [48]. Error metrics, such as mean absolute error (MAE) and root mean square error (RMSE), are typically employed in regression tasks to quantify the difference between predicted and actual values. Nevertheless, RMSE is more suitable for tasks where larger deviations are more critical, as it gives greater weight to larger errors than MAE.

Precision, recall, F1-score, accuracy, receiver operating characteristic (ROC) curve, and its corresponding area under the curve (AUC) are commonly employed for classification issues. Accuracy represents the overall correct prediction rate, while the proportion of correctly predicted positive cases among all predicted positives is indicated by precision. Recall indicates the number of actual positives correctly identified by the model. The mean of precision and recall denotes the F1-score. The metric offers a balanced parameter particularly useful in imbalanced class distributions. The trade-off between sensitivity and specificity across different thresholds is represented by the ROC curve, while the AUC summarises its overall performance. Another metric, hit ratio (HR) primarily assesses whether the user's intended selection appears in an N recommendation collection [49]. The metric is beneficial as it can consistently manages partially true recommendation situations [50].

Classification metrics are preferred over error metrics due to their popularity in current recommendation scenarios, where most recommendations are frequently seen as ranking issues [48]. Trattner et al. [48] also suggested "normalised discounted cumulative gain (nDCG)" as a ranking metric in the classification issue, an enhanced version of the "discounted cumulative gain (DCG)" metric. The nDCG considers the relevance and position of an item in the list of recommendations.

Figure 2 illustrates the GRS evaluation. This study adopted more targeted approach in selecting evaluation metrics based on the problem characteristics and model objectives. Specifically, RMSE was employed to emphasise and penalise larger prediction errors, aligning with regression-based evaluation requirements. For classification tasks, precision, recall, and AUC were incorporated, which are particularly informative in scenarios involving class imbalance. Consequently, a well-rounded assessment of system performance across both regression and classification perspectives was achieved.

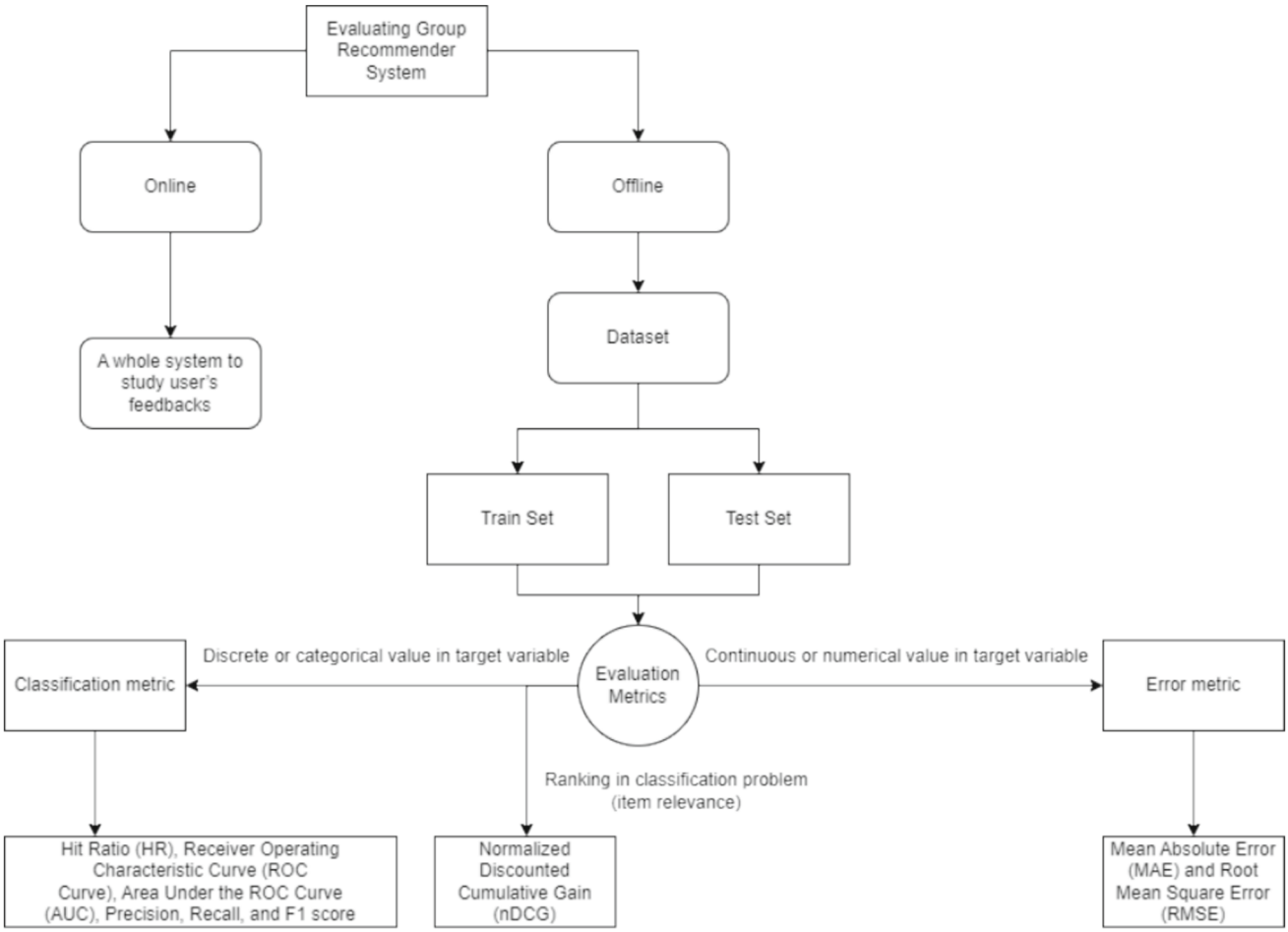


Fig. 2 The GRS evaluation

2.4 Identified research gaps and novel contributions

Despite significant advances in RS research, several limitations remain across individual and group recommendation approaches. Traditional individual RSs often struggle with cold-start and sparsity issues, while CDRS have demonstrated improved performance by transferring knowledge between domains. Likewise, LOD integration has shown promise in alleviating data sparsity for individual RSs. However, research on GRS, particularly cross-domain GRS, remains limited, with few studies exploring LOD integration or the combination of cross-domain and LOD approaches. To systematically highlight these limitations and inform the design of the proposed CDGRS-LOD model, Table 1 provides a comparative overview of existing methods, their datasets, and key challenges.

As summarized in Table 1, existing RS vary widely in methodology, datasets, and performance, yet common limitations persist, including data sparsity, cold-start problems, and limited scalability in group scenarios. These findings underscore the necessity for integrated approaches that leverage both cross-domain knowledge transfer and LOD technologies. The following sections build on this analysis by discussing individual RS, CDRS approaches with and without LOD, and cross-domain GRS, highlighting both their achievements and the persistent gaps that motivate the development of the CDGRS-LOD model.

Individual RS have been widely explored across diverse approaches and datasets. Session-based recommendation systems (SBRS), such as the Capsule GraphSAGE for Session-Based Recommendation (CapsGSR), capture complex item transitions within sessions using both local and global graph structures, achieving notable improvements on Yoochoose and Diginetica datasets [51]. Hybrid recommendation methods for scientific articles combine collaborative filtering (CF) and content-based approaches to mitigate cold-start and sparsity issues, evaluated on CI&T Deskdrop and CiteULike-t [52]. Comparative studies of CF techniques, including Singular Value Decomposition (SVD) and neural CF, highlight differences in ranking and recall on CI&T Deskdrop datasets [53]. Contextual recommendation models using dynamic graph attention networks with adaptive edge

Table 1 Comparative summary of individual and group recommender systems

Methodological category	Existing studies (model/dataset)	Key limitations
Individual RS (Non-CDRS & Non-LOD)	[51]: CapsGSR/Yoochoose, Diginetica [52]: Hybrid RS/CI&T Deskdrop, CiteULike-t [53]: SVD & Neural CF/CI&T Deskdrop [54]: DGAT-AEA/Yelp-NC, Yelp-OH, Amazon-Book [55]: Heterogeneous Graph Attention SRS/Ciao, Epinions, Yelp, Amazon_Fashion	(1) Cold-start and sparsity (2) Limited cross-domain integration (3) Reduced diversity and coverage
CDRS for Individual (without LOD)	[56]: D-GDL/Non-Euclidean 3D object datasets [57]: Parallel GNN with Mutual Regularization/Real-world datasets	(1) Limited handling of domain-specific preferences (2) Reduced effectiveness when domain overlap is low
CDRS for Individual (with LOD)	[38]: CD-SemMF/Source & target domain item metadata [58]: Semantic Similarity Measures/Real-world datasets	(1) High computational complexity (2) Challenges with dynamic user preferences
CDRS for Group (without LOD)	[59]: CDGRS/Tourism datasets (Delhi subdomains) [41]: CDGRS with Generalized Aggregation/CAMRa2011 [60]: HAN-CDGR/Mafengwo, Yelp, MovieLens datasets	(1) Limited dynamic group modeling (2) Cannot incorporate external semantic knowledge
GRS (with LOD, without cross-domain)	[40]: GRS-LOD/MovieLens1M + DBpedia	(1) Data sparsity addressed only at individual level (2) No cross-domain integration (3) Research gap for CDGRS with LOD

attributes (DGAT-AEA) leverage user-item-context relationships to enhance accuracy and novelty on Yelp-NC, Yelp-OH, and Amazon-Book datasets [54]. Social RS employing heterogeneous graph attention networks model multi-type node interactions, addressing cold-start issues and improving prediction on Ciao, Epinions, Yelp, and Amazon_Fashion [55]. Despite these advances, individual RSs remain constrained by data sparsity, extreme cold-start conditions, and limited integration of multi-domain knowledge, which can reduce recommendation diversity and coverage.

CDRS without LOD leverage knowledge transfer across domains to overcome traditional RS limitations. The D-GDL method [56] supports accurate 3D object recommendations in non-Euclidean domains, addressing cold-start and sparsity challenges. Wang et al. [57] applied parallel graph neural networks with mutual regularization to capture both domain-specific and shared user preferences, demonstrating improved performance and interpretability on real-world datasets. While these methods enhance cross-domain recommendation, they still face difficulties in fully capturing domain-specific preferences and maintaining robust performance when domain overlap is limited.

CDRS with LOD further improve recommendations by exploiting semantic relationships across domains. CD-SemMF [38] utilizes item metadata from source and target domains to address cold-start issues in CF, while Wang et al. [58] show that semantic relatedness measures can generate relevant recommendations even in sparse user scenarios. Although effective, these methods involve higher computational complexity and may struggle to adapt to rapidly changing user preferences, highlighting the need for efficient LOD integration strategies.

Cross-domain GRS without LOD address challenges unique to group recommendations. Richa and Bedi [59] introduced a cross-domain GRS framework to reduce cold-start and data sparsity in tourism datasets, outperforming traditional GRS using metrics such as F-measure, precision, recall, and MAE. Liang et al. [60] extended this with HAN-CDGR, a hierarchical attention neural network using adversarial learning to enhance cross-domain knowledge transfer. Experiments on Yelp, CAMRa2011, MovieLens1M,

MovieLens25M, and MovieLens-Simi demonstrate superior performance over multiple baselines. Nevertheless, existing cross-domain GRS approaches are limited in modelling dynamic group preferences and integrating external semantic knowledge.

GRS using LOD (GRS-LOD) aim to alleviate individual-level sparsity before group formation. Nawi et al. [40] proposed a GRS-LOD model incorporating DBpedia to enrich item information, improving prediction accuracy and recommendation relevance on MovieLens1M (ML1M). Aggregation strategies included Average, Most Pleasure, Average without Misery, and Least Misery. While effective, these methods do not integrate cross-domain knowledge, revealing a significant research gap for combining LOD and cross-domain approaches in group recommendations.

Building on these findings, our prior study [34] highlights the untapped potential of integrating LOD and cross-domain approaches in GRS. Previous research has applied CDRS for groups or LOD in GRS separately, but no studies combine both in GRS. Leveraging LOD can mitigate data sparsity, while cross-domain techniques facilitate knowledge transfer across domains. Integrating these strategies can enhance recommendation accuracy and relevance, particularly in scenarios such as collective travel planning or shared entertainment. Future research should explore algorithmic frameworks combining deep learning, matrix factorization, and graph-based methods to integrate cross-domain knowledge with LOD, thereby developing robust and scalable group recommender systems.

3 The CDGRS-LOD model

3.1 Cross-domain feature extraction

This study aimed to enhance group recommendation capabilities by leveraging external data sources and advanced modelling techniques. The methodology employed to develop and evaluate the proposed cross-domain group recommender system (GRS) leveraging linked open data (LOD) (CDGRS-LOD) model is outlined in this section of the article. The methodology is divided into data collection, feature extraction, model development, and evaluation metrics. During data collection, LOD-enriched datasets were gathered, yielding a wealth of contextual information relevant to user preferences and item characteristics. The incorporation of LOD also enabled the model to access structured data that enhanced its understanding of relationships among users and items, addressing the common issue of data sparsity in the recommender systems (RS).

Feature extraction was crucial in the methodology applied in this study. The phase involved identifying and processing relevant attributes from the original dataset and the enriched LOD. The step also ensured the efficiency of the model in capturing the nuances of user preferences and item features, facilitating improved recommendations accuracy.

In the model development phase, the architecture of the CDGRS-LOD model was designed by integrating cross-domain recommendation techniques. Insights from multiple domains were utilised to improve the performance of the model in predicting user preferences for groups, recognising the complexities involved in balancing diverse interests within a collaborative context. Overall, the methodology employed aimed to address the challenges of traditional recommendation systems. The procedure may also contribute to the advancement of group recommendation techniques in an increasingly complex and data-rich environment.

This study extracted a single cross-domain feature with LOD to improve group recommendation performance. The concept was extended by utilising the LOD method to extract all three crucial features, including the cross-domain feature, music composer. The features significantly improved the accuracy of the recommendations and the performance of individual RS when incorporated into the recommendation system [61].

The integration of multiple features, including starring, director, and music composer, enhanced the system's ability to handle complex user preferences across diverse domains. Data sparsity was also further reduced. Furthermore, a more comprehensive and accurate group recommendation outcome was obtained by employing the extracted features and LOD, including the cross-domain feature. Figure 3 demonstrates the connection between the domains based on LOD through DBpedia. A structured knowledge was obtained, revealing how entity relationships across different domains can be utilised for knowledge transfer in RS.

The aim of a cross-domain GRS is to recommend movies to a group of users by establishing patterns within multiple domains. A simulation of multi-domain preferences was created with genres and movie features (such as directors and actors) as proxies for varying "domains". The LOD technology was also applied to extract features from music domain to determine the patterns relevant to the movie dataset. Subsequently, the patterns that can be utilised in movie GRS was transferred. The summary of the proposed model is illustrated in Fig. 4.

Algorithm 1 in Fig. 5 illustrates the pseudo-code for cross-domain feature extraction using LOD from DBpedia, with the goal of enriching a movie dataset with music composer information. The process begins with two inputs: the MovieLens dataset (D) and a mapping file (F) that associates movie titles with their corresponding DBpedia Uniform Resource Identifiers (URIs). The algorithm first creates a duplicate dataset, D′, and adds two attributes, MusicComposer and DBpediaURI.

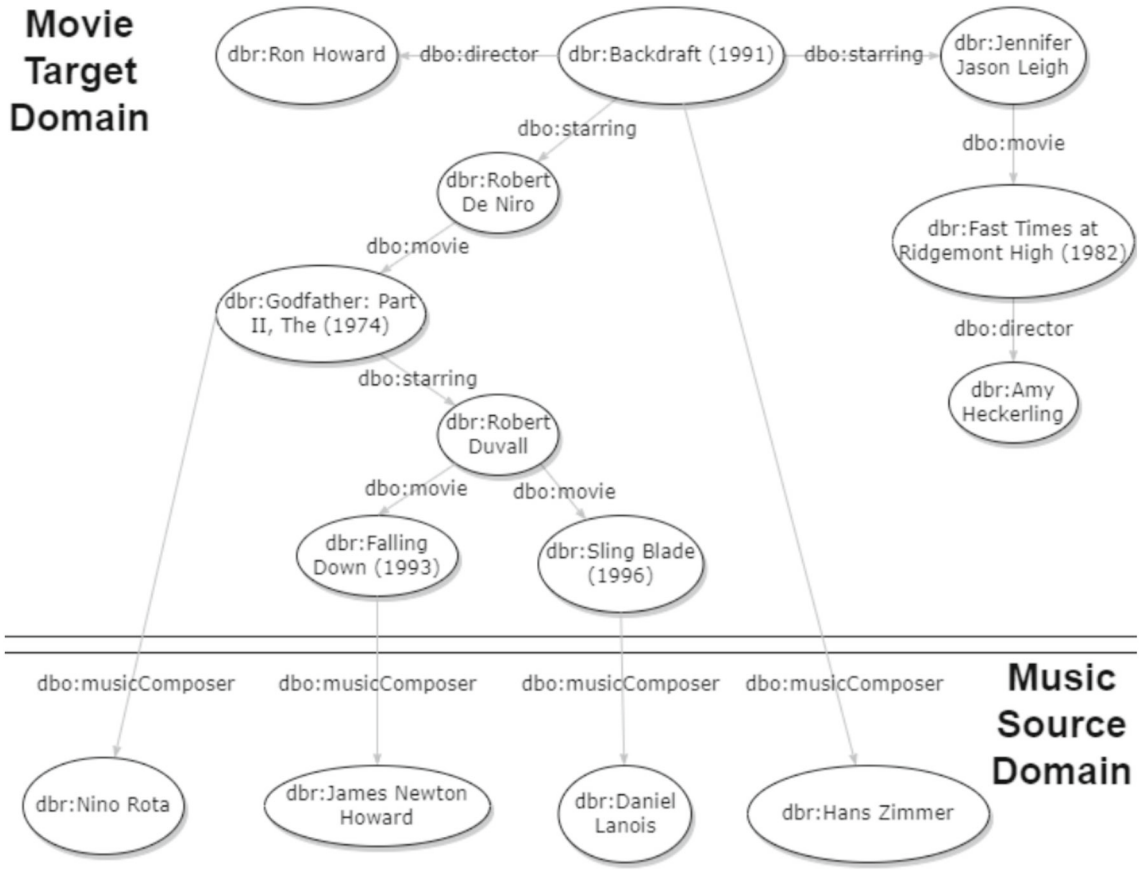


Fig. 3 Selected LOD linked to DBpedia

Each movie record in D is then checked against F to determine whether a DBpedia URI exists. If found, the URI is linked to the movie and added to a query list. A SPARQL query is formulated to retrieve the composer's name in English, and parallel processing is employed to execute multiple queries efficiently across all movies in the list.

For each successful query, the composer name and corresponding URI are stored in D′. The enriched dataset D′ is then saved and returned as the output, now containing the music composer feature for each movie. This process effectively integrates semantic data from the music domain into the movie dataset, enhancing its cross-domain richness and supporting improved recommendation quality.

3.2 Group formation and modelling for recommender system

Group formation is vital in developing a GRS, involving clustering users into groups based on their shared characteristics or preferences. Groups can be established with clustering techniques, such as the k-nearest neighbours (kNN) and K-means. Selecting relevant neighbours in clustering is also critical for improving collaborative filtering (CF) performance [62].

Forming groups with kNN technique for this study was initiated with data preparation, which encompassed user data, such as demographic details, past behaviours, and preferences. This technique automatically detects clusters of homogeneous users [63]. The information was collected and represented in a structured format. A user-item matrix is a common choice, where the users are represented in rows, columns denote items, and cell values indicate user interactions, such as ratings or purchases. Proper preprocessing is also essential, including handling missing values and feature normalisation to ensure consistency.

During group formation, users were clustered based on their similarities. In this study, kNN was employed as the clustering technique. The approach is an effective clustering method for forming groups and allowing users to be duplicated in more than one group [64]. Following the application of the clustering algorithm, the users were partitioned into distinct groups, forming the basis for tailored recommendations.

Figure 6 demonstrates algorithm 2 applied in this study. According to the figure, the group recommendation system

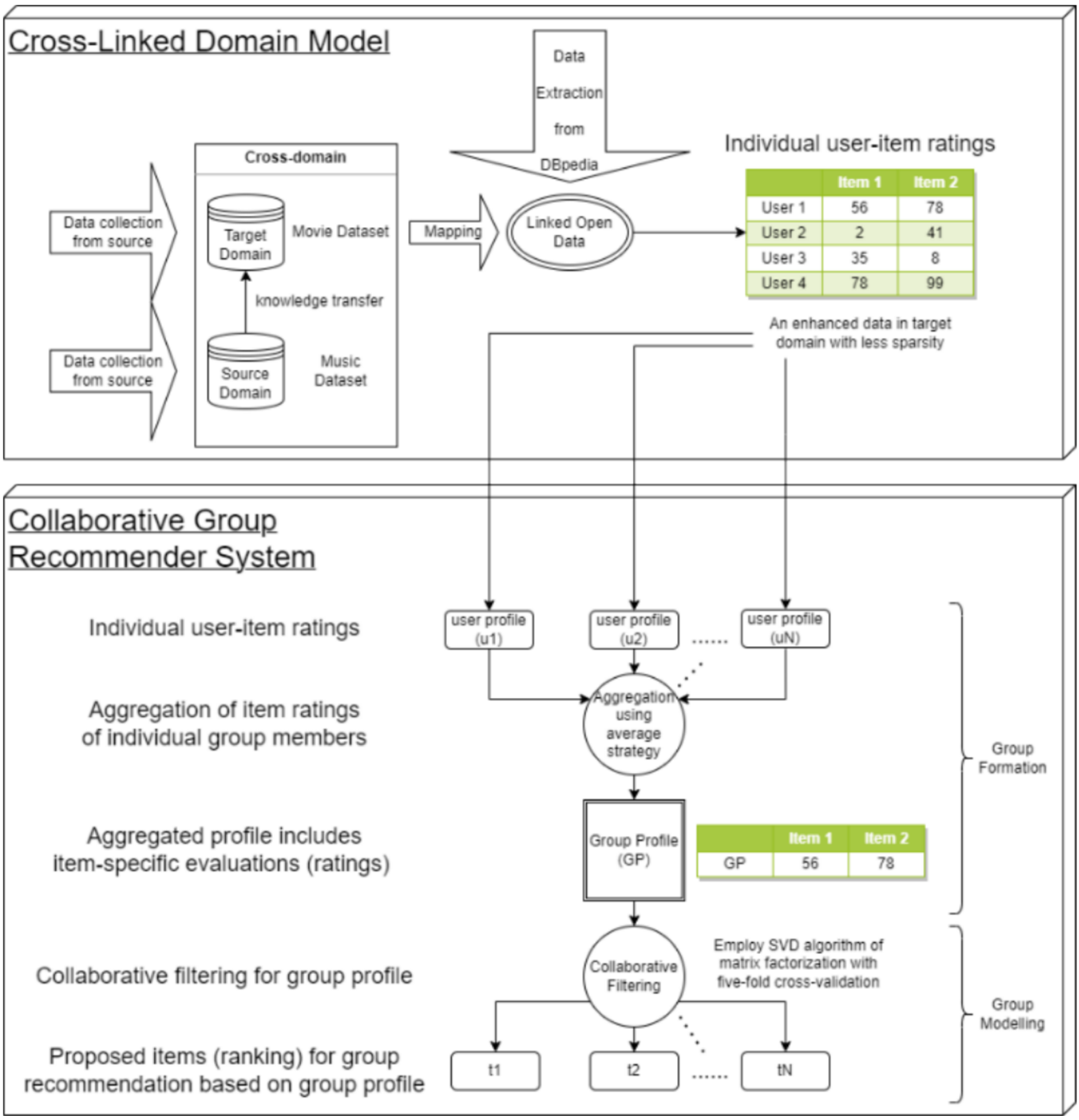


Fig. 4 CDGRS-LOD model

employed singular value decomposition (SVD) and kNN. The algorithm began with a user-movie rating dataset, which comprised UserID, MovieID, and Rating columns. A rating between 1 and 5 was applied. The algorithm also takes the dataset, a specific target user ID, and the desired group size, k, as input. Group size is the number of members in each group.

In this study the data preparation step included loading and preprocessing the information, which resulted in retaining only the essential columns: UserID, MovieID, and Rating. The Reader class in the Surprise library was employed to define the rating scale and converted the dataset into a format compatible with Surprise. The step ensured that the data were ready for subsequent processing and analysis.

The obtained dataset was split into an 80–20 training and test set ratio. The split facilitated the performance evaluation of the model and prevented overfitting. The training set was

Fig. 5 Algorithm 1: cross-domain feature extraction adopting DBpedia

```
Algorithm 1: Cross-Domain Feature Extraction
using LOD from DBpedia
Input: Movie dataset D, Mapping file F
(MovieLens-DBpedia URIs)
Output: Enriched dataset D' with Music Composer
feature

Begin
1. Create a copy of dataset D as D'
2. Add two empty attributes to D': MusicComposer
and DBpediaURI
3. For each movie m in D do
4.     If m exists in mapping file F then
5.         Retrieve the corresponding DBpedia
URI u
6.         Add the pair (m, u) to QueryList
7.     End If
8. End For
9. Define SPARQL query (QUERY) to retrieve the
music composer for each movie where the label
language is English
10. Perform parallel processing with P processes
11.     For each (m, u) in QueryList do
12.         Replace placeholder in QUERY with
the specific DBpedia URI
13.         Execute the SPARQL query
14.         If composer names are found then
15.             Assign the composer names to
D'.MusicComposer[m]
16.             Assign the URI u to
D'.DBpediaURI[m]
17.         End If
18.     End For
19. End Parallel
20. Save the enriched dataset D'
21. Return D'
End
```

employed to train the SVD algorithm, which learns latent factors for users and items, and the kNN algorithm, configured with cosine similarity and user-based recommendations. The kNN model also computes similarity between users, which is essential for forming groups.

The group formation process in this study involved defining a function to identify the top kNN for a given target user based on the kNN model. Firstly, the target user's ID was converted into the internal representation utilised during training before applying the kNN model to establish the nearest neighbours. Subsequently, the internal user ID of the neighbours were mapped back to their original IDs. The algorithm employed in this study can retrieve the top 50 nearest neighbours of a target user to form a group.

This study defined functions to extract latent factors for users and items from the trained SVD model to predict group ratings. A group profile was then computed by aggregating the latent factors of its members, typically by taking their mean. The group profile was utilised to predict the rating for a specific movie by calculating the dot product with the latent factors of the movie. The predicted rating of the group for a movie can also be determined through the approach.

Fig. 6 Algorithm 2: GRS using SVD and KNN

```
Algorithm 2: Group Recommendation System using
SVD and KNN
Input: Dataset, target_user_id, k (group size),
movie_id
Output: Predicted group rating, RMSE, Accuracy

Begin
1. Load dataset and preprocess
2. Define rating scale and prepare data for
Surprise library
3. Split dataset into trainset and testset
4. Train SVD on trainset
5. Train KNN with cosine similarity on trainset
6. Define function get_top_k_neighbours(user_id,
k)
7.     Convert user_id to internal ID
8.     Find k-nearest neighbours using KNN
9.     Convert neighbours back to raw IDs
10.    Return group_user_ids
End Function
11. Define function get_user_latent_factors(svd,
user_id)
12.    Return latent factors for user_id
End Function
13. Define function get_item_latent_factors(svd,
item_id)
14.    Return latent factors for item_id
End Function
15. Define function
predict_group_rating(group_user_ids, movie_id)
16.    Compute mean of latent factors for
group_user_ids
17.    Compute dot product with latent factors
of movie_id
18.    Return predicted group rating
End Function
19. Group Formation:
20.    group_user_ids =
get_top_k_neighbours(target_user_id, k)
21. Group Prediction:
22.    predicted_rating =
predict_group_rating(group_user_ids, movie_id)
23. Evaluate SVD:
24.    Compute RMSE on testset
25.    Compute accuracy based on threshold
26. Output: predicted_rating, RMSE, Accuracy
End
```

Accuracy metrics, such as root mean square error (RMSE), was applied to evaluate the performance of the proposed model. Actual ratings were compared with predicted ratings to determine RMSE, while accuracy was established by setting a threshold, such as 3.5, to classify whether a movie was "liked". The correct prediction percentage was then computed. The output of the algorithm included the predicted group rating, RMSE, and accuracy values of a movie.

In summary, the algorithm developed in this study loaded and pre-processed the dataset, split it into training and testing subsets, trained the SVD and kNN models, formed a group of the kNN, computed the group's profile, predicting group ratings for specific movies, and evaluated the accuracy of the model. The structured approach contributed to accurate and personalised group recommendations.

3.3 Generating recommendations

In this study, recommendations for users within a group were derived from the trained kNN model. For any given user, the model identified their kNN within the group before aggregating recommendations from the neighbours to provide personalised suggestions. SVD was then employed to refine the recommendations. The matrix factorisation technique reduced the dimensionality of the user-item matrix and uncovered latent factors that improve recommendation accuracy.

Matrix factorisation can enhance CF algorithms [65]. SVD is a common technique in CF and can be applied for group recommendations. The technique decomposes the user-item interaction matrix into latent factors, which can then be utilised to predict missing ratings. For group recommendations in this study, SVD was adapted by aggregating individual preferences into a collective group profile.

3.4 Evaluating recommendations

Evaluating a system is critical for determining effectiveness and ensuring reliability in various tasks. Depending on the problem type and goals, varying evaluation metrics are applicable, including precision, recall, and area under the curve (AUC) for classification tasks. Precision focuses on correctly predicted positive percentage instances among all predicted positives, whereas recall (or sensitivity) establishes the proportion of correctly predicted positive instances out of all actual positives.

The AUC, specifically under the receiver operating characteristic (ROC) curve, determines the ability of a model to distinguish between classes across different threshold settings. The metric is particularly useful for imbalanced datasets. The ROC curve is also widely employed for assessing the performance of classification tasks, specifically in distinguishing between different outcome classes [66].

Metrics, such as RMSE and mean squared error (MSE) are widely utilised for regression problems or continuous prediction tasks. The RMSE is valuable as it quantifies the differences between predicted and actual values while expressing errors in identical units as the original data. The approach squares the errors and establishes the square root, penalising larger deviations more heavily, making it suitable for tasks where significant errors must be highlighted.

This study applied RMSE to penalise larger errors. For classification problems, precision, recall, and AUC provided superior insights into performance, particularly with imbalanced datasets. Consequently, a comprehensive understanding of the performance of the system for regression and classification tasks were ensured. The RMSE mathematical formula as an evaluation metric for regression tasks in this study is represented by Eq. (1), where i represents the index of the observations, $\gamma(i)$ is the actual value for data point i, $\gamma(i)$ denotes the predicted value for data point i, and N is the total number of non-missing data points.

$$RMSE = \sqrt{\frac{\sum_{i=1}^{N} ||\gamma(i) - \widehat{y}(i)||^2}{N}} \tag{1}$$

Equations (2) and (3) were employed to calculate precision and recall values. Meanwhile, the ROC function, which defines the relationship between sensitivity and specificity [66], is represented by Eq. (4). Equation (5) mathematically defines AUC.

$$\boldsymbol{P_u} = \frac{|\boldsymbol{hits_u}|}{|\boldsymbol{recset_u}|} \tag{2}$$

$$\boldsymbol{R_u} = \frac{|\boldsymbol{hits_u}|}{|\boldsymbol{testset_u}|} \tag{3}$$

$$ROC(s) = 1 - F_1(F_0^{-1}(1-s)) \tag{4}$$

$$AUC = \int_0^1 ROC(s)ds \tag{5}$$

$\boldsymbol{P}$ refers to precision, hits denote the number of correctly recommended items for user, $\boldsymbol{recset}$ represents recommended items for user, $\boldsymbol{u}$ is the user. While $\boldsymbol{R}$ denotes recall, $\boldsymbol{testset}$ is the hits of the testing set size, F_0 and F_1 are the cumulative distribution functions of the negative and positive classes, respectively, and s ∈ (0,1) represents the specificity level.

The value of AUC can range from 0 to 1, where 1.0 indicates a perfect classifier, while 0.5 reflects performance equivalent to random guessing. Values below 0.5 suggest a model performing worse than random, which can be corrected by inverting the prediction scores. The formulation

provides a theoretical understanding of classifier performance across all thresholds, complementing the threshold-based insights from precision and recall.

4 Materials and methods

4.1 Experimental setup

This study applied the Singular Value Decomposition (SVD) algorithm for rating prediction and employed a fivefold cross-validation approach to evaluate the proposed cross-domain group recommender system (GRS) leveraging linked open data (LOD) (CDGRS-LOD) model. A total of 15 groups were established using the enhanced dataset, with the group size (k) set to 50. The evaluation was conducted in two contexts:

(i)Comparing the performance of CDGRS-LOD under 1% and 5% sparsity reduction levels; and.
(ii)Evaluating the proposed model against existing GRS models, namely GRS-LOD [63], Archi2_bkm [67], Archi2_km [67], and Predict and Cluster [68].

The detailed implementation process, including tools, libraries, and data extraction methods used to develop CDGRS-LOD, is described in the following subsection.

4.1.1 Implementation details

The implementation of the proposed CDGRS-LOD model was conducted using Python. The Surprise library was utilised for implementing the SVD and k-nearest neighbours (kNN) algorithms to perform collaborative filtering (CF) and group formation. The scikit-learn framework was employed for data preprocessing and similarity computation, ensuring model compatibility and efficient handling of large-scale datasets. The SPARQLWrapper library was used to extract LOD from DBpedia, enabling the retrieval of external semantic information such as Music Composer, Director and Starring attributes to enrich item metadata. Parallel processing techniques were integrated to optimise query execution performance during data enrichment. The overall system was executed in a controlled Python environment, and the implementation results were evaluated using the defined experimental settings and metrics.

4.2 Dataset

This study utilised the MovieLens 1 Million (ML1M) dataset to assess the efficacy of the proposed model. Cross-domain and LOD were also applied to enrich the dataset with supplementary movie-related information from DBpedia. DBpedia is a structured LOD platform interconnecting movies with their relevant metadata across multiple domains.

Two interconnected domains, movies and music, were assessed in this study to enhance the recommendation process. The LOD technology was applied to extract additional attributes, such as director and starring, considering the limited data availability within the movie target domain. The step provided deeper contextual insights into parallel movie domains. Furthermore, similar movies were connected with music domain, featuring the music composer. The enriched dataset facilitated a more comprehensive understanding of relationships within the recommendation framework, improving the accuracy and relevance of the proposed approach.

4.3 Aggregation approach in group recommender system

A preferences aggregation approach frequently utilised in GRS [69] was employed in this study. Average strategies were also implemented for the aggregation strategies introduced by Masthoff [70]. This strategy was more democratic, as all group members were treated equally, providing an average score of all group members' ratings [70]. The average strategies have also been extensively utilised in several studies [63, 68, 71]. The average strategies as defined in Eq. 6 was applied in this study, where $Grel(G, i)$ represents the group preferences for item i, while rating of user u for item i referred by Rel_{ui}, and G represents the group cluster.

$$Grel(G, i) = \frac{\sum_{u \epsilon G} Rel_{ui}}{|G|} \tag{6}$$

4.4 The metric

Root mean square error (RMSE), precision, recall and receiver operating characteristic (ROC) curve via area under the curve (AUC) value were employed in this study to evaluate the performance of the proposed model in different sparsity reduction percentages. The RMSE also enabled performance comparisons between the proposed model against existing approaches. The metric served as a reliable parameter for quantifying the accuracy of recommendations, allowing clear comparisons between the proposed model and baseline methods. Relevancy metrics were also applied for precision and recall to determine the relevance of the recommended items to groups, particularly the how accurately the recommended items match the interests of the group.

5 Results and discussion

In this section, the influences of dataset sparsity reduction in the effectiveness of the recommendation are discussed. Experimental results obtained from assessing the proposed model against existing approaches are also highlighted. The evaluations permitted assessments of performance of the models, focusing on the effects of cross-domain recommendations with linked open data (LOD) enrichment on model accuracy and relevancy.

This study analysed key performance indicators, including accuracy, sparsity reduction, and comparison with baseline models in detail. The findings highlighted the benefits of integrating external knowledge sources, such as LOD, and the significance of cross-domain features in addressing sparse data challenges. A thorough results comparison demonstrated the superiority of the proposed cross-domain group recommender system (GRS) leveraging LOD (CDGRS-LOD) model. The proposed model outperformed existing models regarding root mean square error (RMSE), underscoring the importance of advanced techniques in GRS. Figure 7 illustrates the linked data visualisation through Neo4j.

5.1 Results of proposed CDGRS-LOD relative to sparsity reduction

Based on the findings, the proposed CDGRS-LOD model performed better than existing models. The proposed model recorded 0.5917 as the lowest RMSE value, suggesting that applying LOD technology and cross-domain significantly improves its predictive capabilities, particularly by addressing sparsity in the dataset. Reducing 5% sparsity from 95.53 to 90.53% also improved the model, enabling it to manage incomplete or inconsistent data more effectively. The results as shown in Table 2 are based on experiments on a group of 50 with the enhanced dataset.

The quality of recommendations generated for groups is significantly compromised by data sparsity, a persistent and critical challenge in both recommender systems (RS) and GRS. Nevertheless, several steps can simultaneously reduce data sparsity and enhance RS and GRS performance, including cross-domain and LOD. In current study, only cross-domain recommender systems (CDRS) for groups and CDRS with LOD technology for individual RS were implemented, not both.

Although several positive empirical reports are available on the integration of LOD and cross-domain approach in individual RS, the integration of LOD and cross-domain approach in GRS to increase the performance of group recommendations has yet to be established. No related studies on cross-domain GRS with LOD technology currently exist. Consequently, the results obtained in this study provided a notable advancement in this field of study.

The experimental findings documented in this study indicated that the proposed CDGRS-LOD model significantly outperformed other models, recording an RMSE of 0.5917. Furthermore, the robustness of CDGRS-LOD was evidenced by the superior accuracy achieved over existing models and the new benchmark established for GRS.

Several key factors contributed to the superior performance of the proposed model. The integration of LOD into the RS allowed for the inclusion of rich, diverse, and external knowledge sources, enriching user profiles and item representations. Technology also allowed the model to make more informed predictions by leveraging external, structured data that improved the accuracy of group recommendations. The step is particularly vital in scenarios involving sparse primary available data or insufficient context.

Figure 8 demonstrates the performance evaluation of recommendation with a receiver operating characteristic (ROC) curve. The ROC curve, represented in blue, plots the true positive rate (TPR) against the false positive rate (FPR), providing a visual representation of the ability of the proposed model to distinguish relevant and non-relevant recommendations. Meanwhile, the diagonal red dashed line denotes the performance of a random classifier, serving as a baseline for comparison.

The area under the curve (AUC) values range between 0 and 1, where 1 represents a perfect classifier and 0 indicates an inaccurate classifier [66]. The 0.8095 AUC value in this study revealed that the proposed model performed significantly better than random guessing, demonstrating a notable predictive capability. A 73.78% accuracy further validated the reliability of the recommendation system. The findings suggested that the proposed recommendation approach can effectively predict user preferences with considerable precision and recall, making it a suitable method for real-world applications.

The cross-domain recommendation significantly contributed to the success of the developed CDGRS-LOD model. Insights from multiple domains enabled the proposed model to overcome the limitations of a single-domain approach, where data may be too specific or narrow. Diversity and recommendation quality were also enhanced as cross-domain systems utilise knowledge from one domain to improve recommendations in another, leading to more robust and accurate group recommendations. The attribute is particularly effective in group recommendations, where the preferences of multiple users require balancing and comprehension in a broader context.

Figure 9 provides insight into the classification performance of the developed model with 1% sparsity reduced. The 0.8418 AUC value indicates the substantial predictive capability of the recommendation system. Furthermore, the

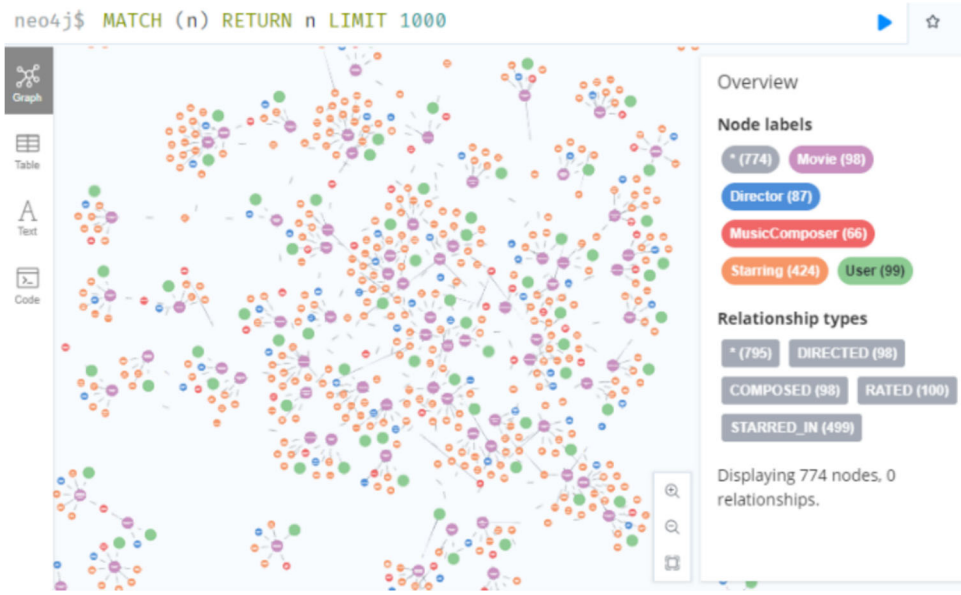


Fig. 7 Linked data on features selected in Neo4j platform

Table 2 Result comparison of proposed CDGRS-LOD based on sparsity reduction

		RMSE	Precision	Recall	AUC (ROC)
1.	Proposed CDGRS-LOD Model (5% Sparsity Reduced)	**0.5917**	**0.8294**	**0.8387**	**0.9034**
2.	Proposed CDGRS-LOD Model (1% Sparsity Reduced)	0.7852	0.7844	0.7969	0.8418
3.	Original dataset with extracted LOD features	0.8746	0.7666	0.7892	0.8095

Bold values indicate the best-performing results for each metric. For RMSE, the lowest value represents the best performance, while for Precision, Recall, and AUC (ROC), the highest values represent the best performance

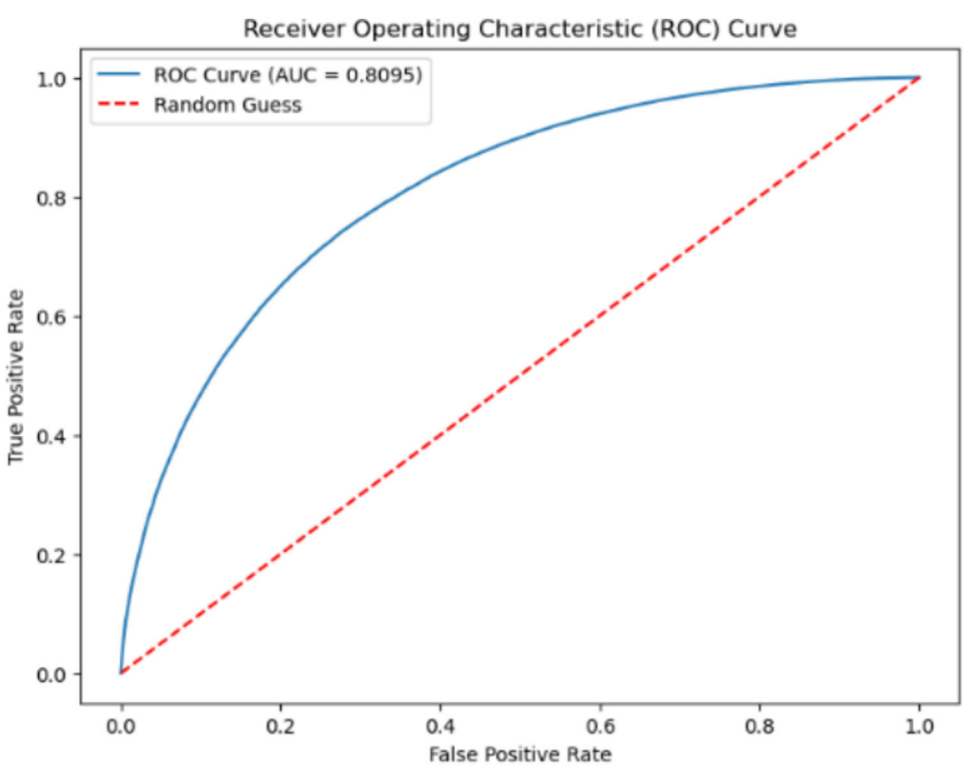


Fig. 8 Performance evaluation using the ROC curve based on original dataset with extracted LOD features

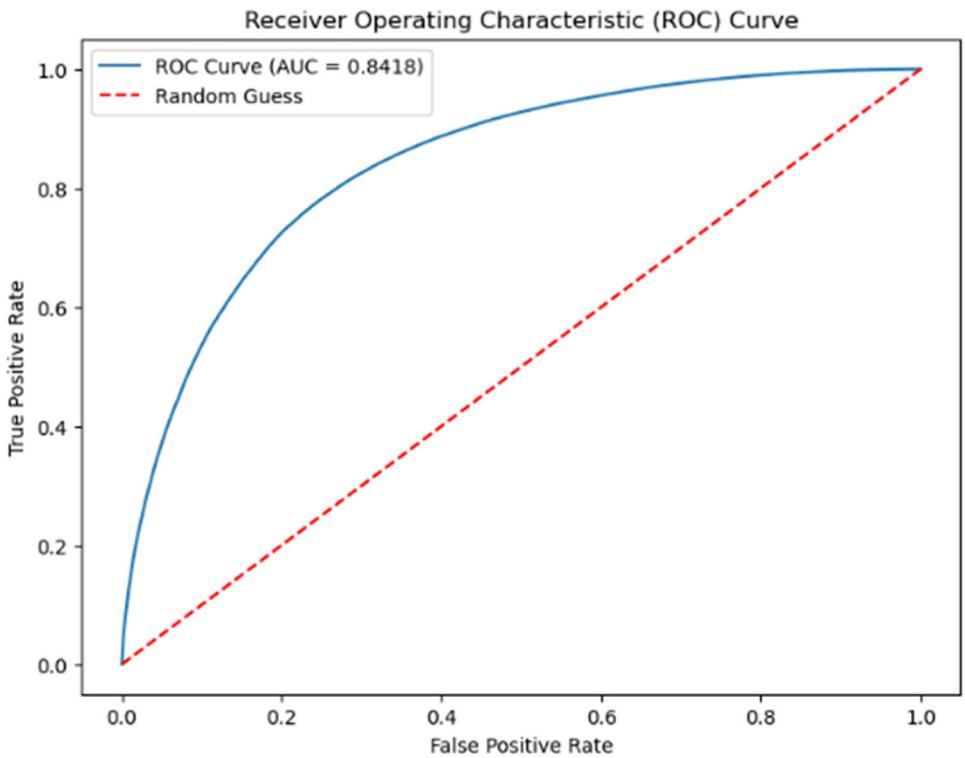


Fig. 9 Performance evaluation using the ROC curve with 1% sparsity reduced

data indicated the effectiveness of the model in distinguishing relevant and non-relevant recommendations.

The 0.9034 AUC value in Fig. 10 demonstrates significant classification performance of the recommendation system, which outperformed the baseline model considerably. The exceptional precision and recall values further reinforce the ability of the system to provide relevant recommendations while maintaining a low FPR. Overall, the results highlighted the robustness and effectiveness of the proposed recommendation approach, a viable method for real-world recommendation tasks.

Sparsity reduction is vital in improving model performance. Decreasing 5% data sparsity from 95.53% to 90.53% affected the effectiveness of the recommendation system. The procedure also allowed more comprehensive user-item interactions. Furthermore, sparsity reduction enhanced the ability of the proposed model to generalise across users and items, increasing prediction accuracy. The superior performance of the CDGRS-LOD model also highlighted the importance of addressing sparsity in RS, particularly when involving group recommendations where individual preferences may not always be well-represented.

5.2 Comparison results of the proposed CDGRS-LOD with existing studies

This study compared the proposed approach to GRS-LOD [63], Predict and Cluster [68], Archi2_bkm [67], and Archi2_km [67]. The developed CDGRS-LOD model was employed with 5% sparsity reduction for the comparison. Based on the results in Table 3, the proposed model outperformed the four existing approaches in terms of efficiency, achieving an RMSE of 0.5917, which represents approximately 24.3% improvement over the best baseline (RMSE = 0.7816).

Table 3 RMSE result of the proposed CDGRS-LOD with existing approaches

	Approaches	RMSE
1	**Proposed CDGRS-LOD Model**	**0.5917**
2	GRS-LOD Model [40]	0.7816
3	Predict & Cluster [68]	0.9435
4	Archi2_bkm [67]	0.9785
5	Archi2_km [67]	0.9781

Bold values indicate the best-performing results for RMSE, where a lower value represents better performance

The Predict & Cluster approach is employed by Boratto and Carta [68] to cluster users into groups using the K-means clustering technique. The K-means method is also employed in the study conducted by Hammou et al. [67], which introduces the Archi2 km and Archi2 bkm (Bisecting K-means). Predict & Cluster approach employs the user-based collaborative filtering (CF) technique to predict items. User-based

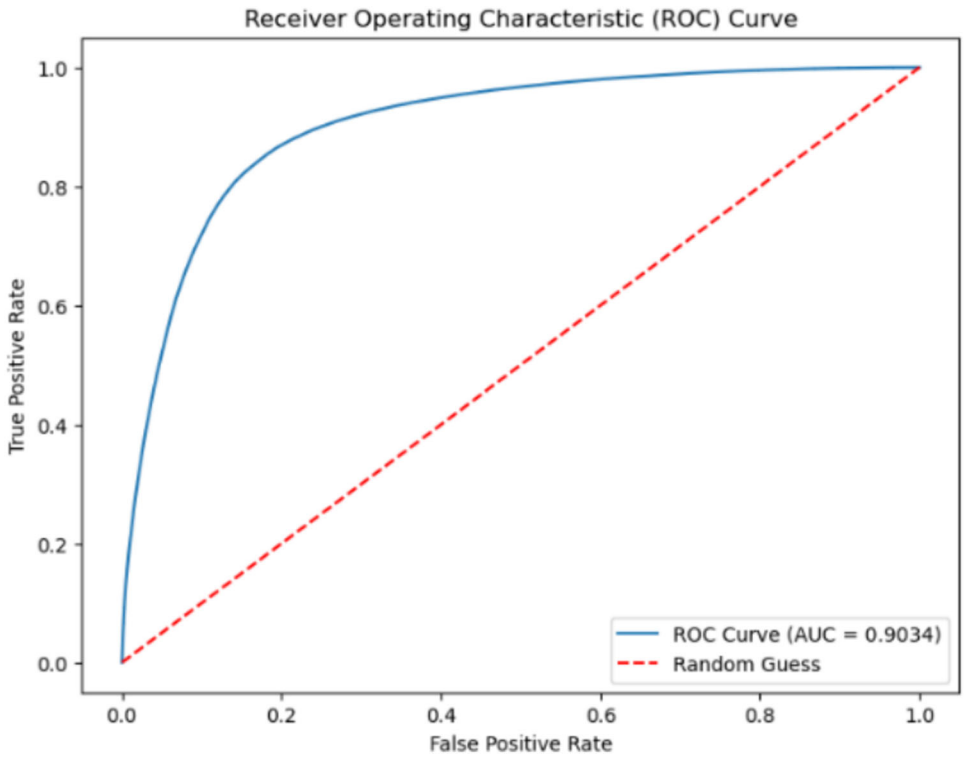


Fig. 10 Performance evaluation using the ROC curve with 5% sparsity reduced

CF is a memory-based RS approach that generates recommendations from the entire database. Makrani [72] found that memory-based RS techniques struggle to handle big data sets. The complexity of each iteration in the rating matrix has an impact on the clustering process for item prediction. In contrast, Archi2_bkm and Archi2_km approach utilized the Apache Spark platform to implement their study using distributed technology, XGBoost (Extreme Gradient Boosting). Through this two-model approach, Hammou et al. [67] evaluate the efficacy of distributed technology that is founded on Big Data on-time computation in the production of recommendations for their work.

Although the GRS-LOD model by Nawi et al. [40] utilised LOD, it did not apply the cross-domain recommendation approach. The model achieved an RMSE of 0.7816. Consequently, integrating LOD is beneficial, the results suggested that the feature was not enough to thoroughly address the complexity of group recommendations without leveraging cross-domain interactions. The gap between the model and the CDGRS-LOD underscored the importance of cross-domain knowledge transfer in boosting recommendation accuracy, specifically when user preferences span across different contexts or domains.

Based on Fig. 11 and Table 3, the traditional models Predict & Cluster, Archi2_bkm, and Archi2_km recorded higher RMSE values, ranging from 0.9435 to 0.9785, than the model proposed in this study. The results demonstrated the limitations of the approaches when presented with sparse data. Primarily relying on internal dataset features and clustering techniques, the conventional models do not sufficiently capture the richness of the relationships between users and items, especially in group recommendation scenarios. The similar performance of the models, with marginal differences in RMSE, also suggests that their inability to incorporate external knowledge leads to poorer performance, particularly in sparse environments where data is insufficient for accurate predictions.

The four existing approaches do not incorporate LOD technology and cross-domain approaches and demonstrate the limitations of traditional recommendation systems when provided with significantly sparse data. Overall, the comparison highlighted the importance of incorporating external knowledge and cross-domain features, as in the developed CDGRS-LOD model, to improve recommendation system performance, particularly in sparse environments. As stated in [22], the sparsity issues must be resolved prior to the aggregation process in GRS. It consequently influences the clustering process in group formation. The results demonstrate that recommendations are more effective when generated through group formation with the proposed model, which results in homogeneous user groups. Thus, the CDGRS-LOD model was able to mitigate the sparsity and demonstrate that overcoming it could enhance the quality of a group's recommendation in comparison to the other previous study.

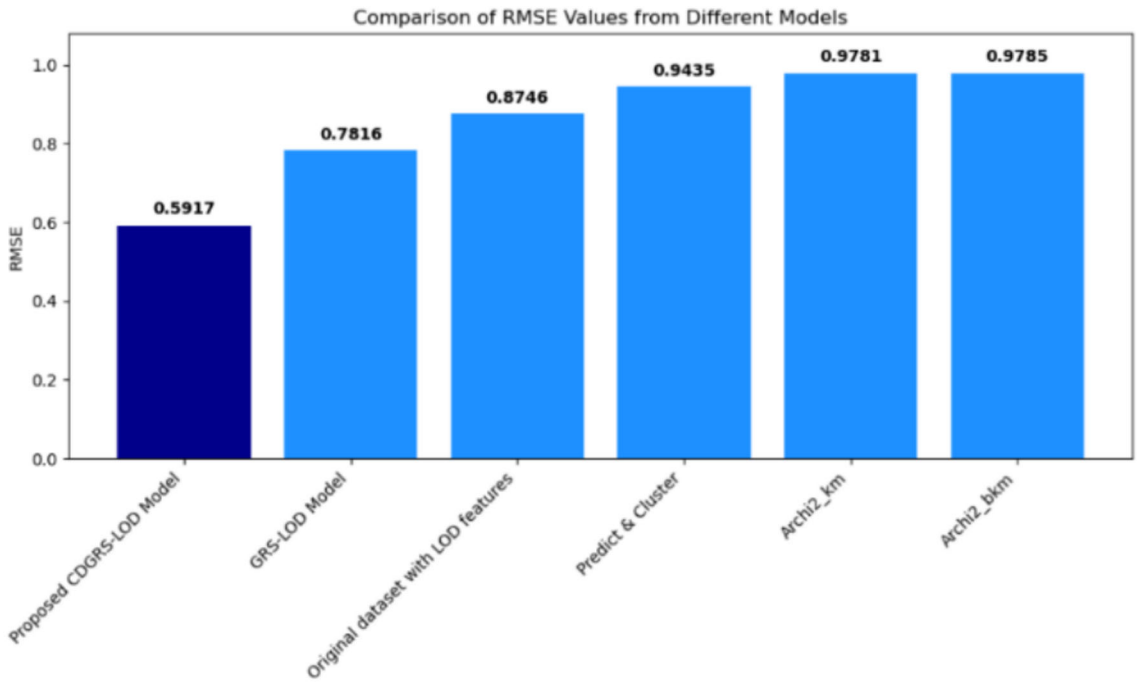


Fig. 11 Graph comparison of RMSE results for the proposed CDGRS-LOD with existing approaches

The CDGRS-LOD model was proposed to provide a comprehensive framework for developing a robust recommendation system that effectively utilises external data and cross-domain insights to yield enhanced group recommendations. According to the comparison and analysis of the results, integrating LOD and cross-domain recommendations in modern GRS is critical. The ability of the proposed CDGRS-LOD model to employ external data sources and cross-domain knowledge allowed it to generate more accurate and relevant recommendations. Consequently, the model outperformed traditional models that are limited by the constraints of a single domain and sparse data. The findings suggested that future studies in recommendation systems should prioritise the inclusion of diverse external knowledge sources and explore cross-domain synergies to elevate their performance in various contexts.

Extending from these findings, the comparative analysis with existing approaches not only validates the technical superiority of the proposed CDGRS-LOD model but also underscores its relevance in addressing real-world group recommendation challenges such as travel planning, event coordination, and collaborative content selection. The lower RMSE value achieved by the model reflects a significant improvement in predictive accuracy, while a higher AUC value, where applicable, indicates enhanced capability in ranking relevant items. These outcomes demonstrate that the integration of LOD with the cross-domain approach contributes to greater precision, relevance, and diversity in group recommendations. Moreover, the model exhibits resilience in sparse data environments, ensuring consistent recommendation quality where limited user-item interactions exist. Future research could further integrate the cross-domain approach within a broader context-aware framework, enabling the system to consider situational and domain-specific factors for more personalised and adaptive group recommendations. This reinforces the potential of the CDGRS-LOD model as a scalable and context-aware framework for practical group recommender system applications.

6 Conclusion and future work

The results obtained in this study affirmed the effectiveness of the proposed cross-domain group recommender system (GRS) leveraging linked open data (LOD) (CDGRS-LOD) model in enhancing the accuracy of group recommendations by integrating LOD with cross-domain approaches. The model also outperformed available methodologies, demonstrating its capability to address data sparsity and significantly improve recommendation quality. Furthermore, data sparsity reduction highlighted the benefits of incorporating external knowledge sources, which enriched the understanding of user preferences and item characteristics.

The ability of the developed CDGRS-LOD model to leverage diverse data streams underscores its potential to deliver more personalised and accurate recommendations and reflects the increasing demand for recommendation systems to adapt to the complexities of user behaviour across

multiple domains. Consequently, this study opens several promising avenues for future enhancement. Firstly, the proposed model can be augmented with additional features and applied to other domains with similar methodological frameworks. This study also laid a foundational step for the GRS field, offering a robust approach to initiate further development and deeper exploration.

Optimising LOD integration can be considered by future works to improve the extraction and representation of relevant features. Adopting advanced machine learning techniques, such as deep learning, may also offer deeper insights and enable more effective feature learning from structured and unstructured data. Moreover, expanding the scope of cross-domain recommendations by incorporating more diverse data sources can elevate the robustness and adaptability of the suggested model in various applications, including personalised marketing and collaborative filtering (CF).

Future studies can develop dynamic, real-time recommender systems (RS) that adapt to evolving user preferences and continually altering datasets. Learning process can be enhanced by incorporating user feedback mechanisms, enabling the system to refine its recommendations based on immediate user interactions. Furthermore, addressing the ethical implications of using external data sources, especially concerning privacy and data security, is crucial, considering the increasingly integrated recommendation technologies into daily digital experiences. The generalisability of the proposed CDGRS-LOD model across different user groups and domains should also be explored. Longitudinal studies may reveal how recommendations influence user behaviour over time, leading to the development of systems that recommend effectively and foster sustained user engagement and satisfaction.

This study provides a significant foundation for enhancing group recommendation systems through the integration of LOD and cross-domain techniques. Future studies can consider refining the approaches and addressing the identified challenges to drive significant advancements towards more intelligent, adaptable, and user-centred recommendation systems that cater to the diverse users' requirements across multiple domains.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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